Q1 - 25 June - Shift 2

$$\lim_{x\to\frac{\pi}{2}}\left(\tan^2 x\left(\left(2\sin^2 x+3\sin x+4\right)^{\frac{1}{2}}-\left(\sin^2 x+6\sin x+2\right)^{\frac{1}{2}}\right)\right)$$

is equal to

(A) $\frac{1}{12}$ (B) $-\frac{1}{18}$

(C) $-\frac{1}{12}$ (D) $-\frac{1}{6}$

Space for your notes:

Q2 - 26 June - Shift 1

$$\lim_{x\to\frac{1}{\sqrt{2}}}\frac{\sin(\cos^{-1}x)-x}{1-\tan(\cos^{-1}x)}$$ is equal to :

(A) $\sqrt{2}$ (B) $-\sqrt{2}$

(C) $\frac{1}{\sqrt{2}}$ (D) $-\frac{1}{\sqrt{2}}$

Space for your notes:

Q3 - 26 June - Shift 2

$$\lim_{x\to 0}\frac{\cos(\sin x)-\cos x}{x^4}$$ is equal to :

(A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{12}$

Space for your notes:

Q4 - 27 June - Shift 1

Let a be an integer such that $\lim_{x\to 7}\frac{18-[1-x]}{[x-3a]}$ exists, where [t] is greatest integer $\leq$ t. Then a is equal to :

(A) -6 (B) -2

(C) 2 (D) 6

Space for your notes:

Q5 - 27 June - Shift 2

Let [t] denote the greatest integer $\leq$ t and {t} denote the fractional part of t. Then integral value of α for which the left hand limit of the function

$f(x)=[1+x]+\frac{\alpha^{2[x]+\{x\}}+[x]-1}{2[x]+\{x\}}$ at x = 0 is equal to $\alpha-\frac{4}{3}$ is ____

Space for your notes:

Q6 - 28 June - Shift 2

The value of $\lim_{n\to\infty} 6\tan\left\{\sum_{r=1}^{n}\tan^{-1}\left(\frac{1}{r^2+3r+3}\right)\right\}$ is equal to

(A) 1 (B) 2

(C) 3 (D) 6

Space for your notes:

Q7 - 28 June - Shift 2

If $\lim_{x\to 1}\frac{\sin(3x^2-4x+1)-x^2+1}{2x^3-7x^2+ax+b}=-2$, then the value of (a – b) is equal to

Space for your notes:

Q8 - 29 June - Shift 2

The value of $\lim_{x \to 1} \frac{\left(x^2-1\right)\sin^2(\pi x)}{x^4-2x^3+2x-1}$ is equal to:

(A) $\frac{\pi^2}{6}$ (B) $\frac{\pi^2}{3}$

(C) $\frac{\pi^2}{2}$ (D) π^2

Space for your notes:

Answer Key

Q1 (A)	**Q2** (D)	**Q3** (C)	**Q4** (A)
Q5 (3)	**Q6** (C)	**Q7** (11)	**Q8** (D)

Q1 (A)

$$\lim_{x\to\frac{\pi}{2}} \tan^2 x\left[\sqrt{2\sin^2 x+3\sin x+4}-\sqrt{\sin^2 x+6\sin x+2}\right]=$$

$$\lim_{x\to\frac{\pi}{2}} \frac{\tan^2 x\,[\sin^2 x-3\sin x+2]}{\sqrt{9}+\sqrt{9}}$$

$$=\lim_{x\to\frac{\pi}{2}} \frac{\tan^2 x(\sin x-1)(\sin x-2)}{6}$$

$$=\frac{1}{6}\lim_{x\to\frac{\pi}{2}} \tan^2 x(1-\sin x)$$

$$=\frac{1}{6}\lim_{x\to\frac{\pi}{2}} \frac{\sin^2 x\,(1-\sin x)}{(1-\sin x)(1+\sin x)}=\frac{1}{12}$$

Q2 (D)

$$\lim_{x\to\frac{1}{\sqrt{2}}} \frac{\sin(\cos^{-1} x)-x}{1-\tan(\cos^{-1} x)}$$

$$\lim_{x\to\frac{1}{\sqrt{2}}} \frac{\sin\left(\sin^{-1}\sqrt{1-x^2}\right)-x}{1-\tan\left(\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)\right)}$$

$$\lim_{x\to\frac{1}{\sqrt{2}}} \frac{\sqrt{1-x^2}-x}{1-\left(\frac{\sqrt{1-x^2}}{x}\right)}$$

$$\lim_{x\to\frac{1}{\sqrt{2}}} (-x)=-\frac{1}{\sqrt{2}}$$

Q3 (C)

$$\lim_{x\to 0}\frac{\cos(\sin x)-\cos x}{x^4};\left(\frac{0}{0}\right)$$

$$\lim_{x\to 0}\left(\frac{2\cdot\sin\left(\frac{x+\sin x}{2}\right)\cdot\sin\left(\frac{x-\sin x}{2}\right)}{x^4}\right)$$

$$\lim_{x\to 0}2\left(\frac{\sin\left(\frac{x+\sin x}{2}\right)}{\left(\frac{x+\sin x}{2}\right)}\right)\left(\frac{\sin\left(\frac{x-\sin x}{2}\right)}{\left(\frac{x-\sin x}{2}\right)}\right)\frac{\left(\frac{x+\sin x}{2}\right)}{x^4}\left(\frac{x-\sin x}{2}\right)$$

$$\lim_{x\to 0}\left(\frac{x^2-\sin^2 x}{2x^4}\right):\left(\frac{0}{0}\right)$$

Apply L-Hopital Rule :

$$\lim_{x\to 0}\frac{2x-2\sin x\cos x}{2.4.x^3}$$

$$\lim_{x\to 0}\frac{2x-\sin 2x}{8x^3};\frac{0}{0}$$: Again apply L-Hopital rule

$$\lim_{x\to 0}\frac{2-2\cos(2x)}{8(3)x^2}$$

$$\lim_{x\to 0}\frac{2(1-\cos(2x))}{24(4x^2)}\times 4\Rightarrow\frac{2}{24}\times\frac{1}{2}\times 4\Rightarrow\frac{1}{6}$$

Q4 (A)

$$\lim_{x\to 7}\frac{18-[1-x]}{[x]-3a}$$

L.H.L. $\lim_{x\to 7-}\frac{18-[1-x]}{[x]-3a}$

$$=\frac{18-(-6)}{6-3a}$$

$$=\frac{24}{6-3a}$$

R.H.L. $\lim_{x\to 7+}\frac{18-[1-x]}{[x]-3a}$

$$=\frac{18-(-7)}{7-3a}$$

$$=\frac{25}{7-3a}$$

Now L.H.L. = R.H.L.

$$\frac{24}{6-3a}=\frac{25}{7-3a}$$

$$\Rightarrow 168-72a=150-75a$$

$$\Rightarrow 18=-3a$$

$$\Rightarrow a=-6$$

Q5 (3)

$$f(x) = [1+x] + \frac{\alpha^{2[x]+\{x\}} + [x] - 1}{2[x] + \{x\}}$$

$$\lim_{x \to 0^-} f(x) = \alpha - \frac{4}{3} \Rightarrow 0 + \frac{\alpha^{-1} - 2}{-1} = \alpha - \frac{4}{3}$$

$$\Rightarrow 2 - \frac{1}{\alpha} = \alpha - \frac{4}{3}$$

$$\Rightarrow \alpha + \frac{1}{\alpha} = \frac{10}{3}$$

$$\Rightarrow \alpha = 3; \alpha \in I$$

Q6 (C)

$$T_r = \tan^{-1}\left[\frac{(r+2)-(r+1)}{1+(r+2)(r+1)}\right]$$

$$= \tan^{-1}(r+2) - \tan^{-1}(r+1)$$

$$T_1 = \tan^{-1} 3 - \tan^{-1} 2$$

$$T_2 = \tan^{-1} 4 - \tan^{-1} 3$$

$$T_n = \tan^{-1}(n+2) - \tan^{-1}(n+1)$$

$$S_n = \tan^{-1}(n+2) - \tan^{-1} 2 = \tan^{-1}\left(\frac{n+2-2}{1+2(n+2)}\right)$$

$$= \tan^{-1}\left(\frac{n}{2n+5}\right)$$

$$\lim_{n \to \infty} 6 \tan\left(\tan^{-1}\left(\frac{n}{2n+5}\right)\right)$$

$$= \lim_{n \to \infty} \frac{6n}{2n+5} = \frac{6}{2} = 3$$

Q7 (11)

$$\lim_{x\to 1}\frac{\sin\left(3x^2-4x+1\right)-x^2+1}{2x^3-7x^2+ax+b}=-2$$

For finite limit

$a+b-5=0$ ...(1)

Apply L'H rule

$$\lim_{x\to 1}\frac{\cos(3x^2-4x+1)(6x-4)-2x}{(6x^2-14x+a)}=-2$$

For finite limit

$6-14+a=0$

$\boxed{a=8}$

From (1) $\boxed{b=-3}$

Now $(a-b)=11$

Q8 (D)

$$\lim_{x\to 1}\frac{(x^2-1)\sin^2\pi x}{(x^2-1)(x-1)^2}=\lim_{x\to 1}\left(\frac{\sin((1-x)\pi))}{\pi(1-x)}\right)^2\pi^2=\pi^2$$